

Algorithm (Pseudocode)

BEGIN

INPUT EntryTime

INPUT ExitTime

INPUT VehicleType

INPUT MembershipStatus

CarRate = 3.0

MotorcycleRate = 1.5

VanRate = 5.0

MembershipDiscountRate = 0.05

// Step 1: Calculate parking duration

ParkingHours = ExitTime - EntryTime

IF ParkingHours < 0 THEN

 ParkingHours = ParkingHours + 24

END IF

ParkingHours = CEILING(ParkingHours)

// Step 2: Determine hourly rate based on vehicle type

IF VehicleType = "Car" THEN

 HourlyRate = CarRate

ELSE IF VehicleType = "Motorcycle" THEN

 HourlyRate = MotorcycleRate

ELSE IF VehicleType = "Van" THEN

 HourlyRate = VanRate

ELSE

```
    OUTPUT "Invalid vehicle type"
    STOP
END IF

// Step 3: Calculate basic parking charge
BasicCharge = ParkingHours * HourlyRate

// Step 4: Apply membership discount
IF MembershipStatus = "Member" THEN
    Discount = BasicCharge * MembershipDiscountRate
ELSE
    Discount = 0
END IF

// Calculate final charge
FinalCharge = BasicCharge - Discount

// Output results
OUTPUT "Parking Duration: ", ParkingHours, " hours"
OUTPUT "Basic Charge: RM", BasicCharge
OUTPUT "Discount: RM", Discount
OUTPUT "Final Charge: RM", FinalCharge

END
```